Fiscal Year:	FY 2019	**Task Last Updated:**	FY 02/27/2019
PI Name:	McLean, Robert Ph.D.		
Project Title:	Polymicrobial Biofilm Growth and Control during Spaceflight		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78666-4684	**Congressional District:**	35
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/05/2016	**End Date:**	12/04/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	October 2017 report: Key personnel on the original grant application are unchanged. This includes co-Principal Investigator (PI) Cheryl Nickerson and her team (Jennifer Barrila and Jiseon Yang) at Arizona State University, as well as C Mark Ott (NASA-Johnson Space Center-JSC). Two collaborators at JSC (Mayra Nelman and Simon Clemett) will be involved primarily on post-flight analysis. Starla Thornhill was a masters' student as the grant was being written. She successfully defended her MS thesis in 2016 and is now doing her PhD on work directly related to this grant. I have one new MS student, Quentin DiPasquale, who is doing modeled microgravity work that is peripheral to the NASA-funded project. As well, we have added Alistair McLean as an unpaid collaborator. He works as a software engineer and is writing a program that will help us enumerate bacterial populations from microscopy images.		
COI Name (Institution):	Barrila, Jennifer Ph.D. (Arizona State University) Ott, C. Mark Ph.D. (NASA Johnson Space Center) Nickerson, Cheryl Ph.D. (Arizona State University)		
Grant/Contract No.:	NNX17AC79G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bacteria and other microorganisms have been shown to grow as surface-adherent biofilm communities in the microgravity environment during spaceflight. Organisms within biofilms are quite resistant to traditional antimicrobial chemicals and can foul water treatment filters resulting in potential risk to the astronauts. The biofilms on International Space Station (ISS) consist of many species, yet studies to date have only tested single-species biofilms. There is growing evidence that mixed-culture biofilms behave quite differently from pure culture biofilms, and we anticipate this phenomenon to occur in the microgravity environment of ISS. Through this study we will investigate the structure by which two-species, mixed-culture biofilms form under microgravity on stainless steel used to construct Environmental Controls and Life Support Systems (ECLSS) on the ISS. As well, we will explore the efficacy by which mixed-culture biofilms can be controlled, and whether microgravity-grown mixed-culture biofilms cause corrosion on stainless steel. Our results are designed to both enable future spaceflight missions and translate into information for the general scientific community on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In work accomplished to date, we have successfully shown that mixtures of different bacteria can grow and form aggregates on surfaces and in liquid (bacterial biofilms), when grown under conditions that mimic the water recycling equipment on the ISS. Our experiments involved using the bacteria Escherichia coli (E. coli) and Pseudomonas aeruginosa (P. aeruginosa). As well, we have just inserted genes that code for fluorescent proteins into each of these organisms so that we can identify and distinguish them using a microscope as E. coli will fluoresce red and P. aeruginosa will fluoresce green. We are also developing a software program that can be used to count and analyze differently stained bacteria in microscopy images. This program is intended to be freely available to the scientific community. These two organisms, E. coli and P. aeruginosa, are often found in water recycling systems and our future work will let us determine whether they need to grow together in a microgravity environment, does one organism protect another if a disinfectant is used. That protection appears to occur in several microbial fouling environments on Earth and one benefit of the proposed work will be to investigate how reduced gravity affects bacterial protection against disinfectants. In future planned work, we will also be doing some tests to see whether the E. coli and P. aeruginosa microorganisms may cause corrosion of stainless steel and if gravity or reduced gravity has an influence on this process.
Task Progress:	In the first year of the grant, we purchased necessary equipment for conducting simulated weightlessness (modeled microgravity) experiments in the lab. In our investigations the plan was to use two representative bacteria, Escherichia coli (E. coli) and Pseudomonas aeruginosa (P. aeruginosa). While the P. aeruginosa strain PAO-1 had been grown successfully in several ground-based modeled microgravity and even flight-based studies, the E. coli strains (originally associated with urinary tract infections) had never been studied, although there is at least one incident (Apollo 13) in which an astronaut had a urinary infection, so the choice of an organism with that infection potential is quite relevant. To this end, we obtained 12 E. coli strains that had been associated with a variety of urinary tract infections from three labs with extensive background in urinary infection research; the University of Michigan ((Principal Investigator (PI) – HLT Mobley)), Baylor College of Medicine (PI – R Hull), and the University of Montreal (PI – J Harel) in Canada. These strains were evaluated for potential growth in a modified artificial urine medium (mAUM) under modeled microgravity (MMG) at ambient temperature (25C), which mimics conditions on the ISS. Based on the ability of these bacteria to grow as biofilms in mAUM in modeled microgravity, we selected our bacterial strains (P. aeruginosa PA01 and E. coli F11). The P. aeruginosa strain contained the gene for the green fluorescent protein (gfp) and we inserted a gene for a red-fluorescing protein (mCherry) into the E. coli F11 strain so that these two bacteria could be distinguished by microscopy as green- and red-fluorescing cells. For year two, we are progressing towards getting flight readiness. Much of our experimental work is being done in concert with Bioserve at the University of Colorado, who is doing the hardware (Biocell) preparation that will be used for flight experiments. Objectives and achievements during the second year of the grant were: 1. Finalized media formulation – Fluids list has been forwarded to Bioserve 2. Finalized Biocell design and cleaning / sterilization protocol – Shared with Bioserve 3. We can get biofilms growing reproducibly on stainless steel in Biocells 4. We can image biofilms, although due to limitations of excitation wavelength of LMM (light microscopy module) microscope, will do all imaging post-flight – Explained below 5. Fluorescent image improved with post-flight processing (adding anti-gfp antibodies) – TBD influence of this processing on structure 6. Worked out ratio of bacteria (Ec-Pa) to send as frozen inoculum – TBD stability of frozen inoculum, which would be relevant with mission scrubs. Work in progress: 7. Stability of fixed biofilm samples in different fixative (paraformaldehyde (PFA)) concentrations 8. Effective concentrations of Ag for biofilm control 9. Selection of sample time points. Details of progress: The media formulation and culture conditions allow us to reproducibly grow biofilms on the Biocell flight hardware devices (objectives 1-4). The Biocells have 12 wells which include a stainless steel coupon, mounted on a pedestal and the chambers are covered with teflon. Two issues that were resolved were: A) the preparation and cleaning of the stainless steel, and B) the final design of the Biocells including the 2 mm distance between the stainless steel coupon and the teflon membrane. Repeated tests have shown that reproducible biofilms are grown in this device (objectives 2

	and 3). For imaging (objective 4), our fluorescent molecules Gfp and mCherry require excitation spectra of 395 nm and 587 nm, respectively. The current laser for confocal microscopy on the ISS light microscopy module (LMM) has a 532 nm wavelength, which would weakly image the mCherry molecule on E. coli F11 but would not detect the Gfp molecule on P. aeruginosa PA01. Confocal microscopy is necessary to remove background light during biofilm imaging as we do get biofilm growth on both the teflon membrane as well as the stainless steel coupons. As a result, we will not use the LMM on the ISS but rather conduct all microscopy post-flight. Work in progress (near completion): We are optimizing protocols to enhance signals using post-flight treatments with fluorescently-tagged antibodies to the reporter molecules Gfp and mCherry (objective 5). As well, we are confirming the protocol to send cultures to launch so as to accommodate potential launch scrubs and associated delays (objective 6). As well, we are testing the stability of biofilms in varying lengths of fixation and concentrations of paraformaldehyde (PFA) (objective 7). Culture and biofilm stability will also be an essential aspect of selecting the time points (objective 9) for imaging during flight.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Thornhill SG, McLean RJC. "Use of whole-cell bioassays for screening quorum signaling, quorum interference, and biofilm dispersion." Methods Mol Biol. 2018;1673:3-24. https://doi.org/10.1007/978-1-4939-7309-5_1 ; PubMed PMID: 29130160 , Dec-2018
Books/Book Chapters	Yang J, Thornhill SG, Barrila J, Nickerson CA, Ott CM, McLean RJC. "Microbiology of the built environment in spacecraft used for human flight." in "Microbiology of atypical environments. Methods in Microbiology. Vol. 45." Ed. V. Gurtler, J.T. Trevors. London: Academic Press/Elsevier, 2018. p. 3-26. https://doi.org/10.1016/bs.mim.2018.07.002 , Oct-2018